

04-638: Programming for Data Analytics

Analytics Assignment III: Individual Assignment

Release Date: 27th November 2025

Due Date: 7th December 2025 - 11:59 PM CAT

Points Total: 100 pts

Assignment Overview

This is an individual assignment that extends your group's work from Assignments I and II. While you will use the same dataset as your group members, you must develop a unique analytical approach that distinguishes your work from your peers. Your analysis will be documented in a Jupyter Notebook, and your findings will be presented in a formal research paper written using Overleaf with the AAAI 2025 conference template.

Critical Requirements:

- You will work with your group's dataset but must ensure your individual work is original and distinct from other group members' submissions
- Your final deliverable includes both a comprehensive Jupyter Notebook with your analysis and a professionally formatted research paper (PDF) written in LaTeX using Overleaf

Part 1: Topic Registration and Differentiation [10pts]

Due: 28th November 2025 - 11:59 PM CAT

Instructions:

- Access the Group Topic Registration Google Sheet [here](#)
- Find your group's section in the sheet
- Register your individual approach by specifying:
 - Your research question/hypothesis
 - Brief description of your unique angle (50-100 words)

Differentiation Requirements:

Your individual work must differ from your group members' work in **at least THREE (3)** of the following ways:

- ☐ **Different research question/hypothesis** - Explore a different aspect of the dataset
- ☐ **Different modelling approach** - Use different algorithms or model types
- ☐ **Different evaluation metrics** - Focus on different performance measures
- ☐ **Different data subset** - Analyse a specific period, region, or category
- ☐ **Different theoretical framing** - Apply different domain perspectives or objectives
- ☐ **Different feature engineering** - Create and utilize different derived features

Important Notes:

- **You CANNOT use any of the three hypotheses from Assignment II** - formulate new ones
- Review your group members' registered topics to avoid overlap

Part 2: Literature Review (15 points)

[10 pts] Background Research:

- Review at least **8-10 relevant sources** (academic papers, reports, credible articles)
- Summarize existing knowledge on your topic
- Identify gaps your analysis will address
- Establish theoretical or practical motivation for your research

[5 pts] Related Work:

- Discuss previous analytical approaches to similar questions
- Explain how your approach differs or builds upon existing work
- Cite sources properly using a consistent citation style (aaai2026)

Part 3: Data Analysis (35 points)

[10 pts] Exploratory Data Analysis:

- Comprehensive data exploration and visualization
- Statistical summaries and distributions
- Identification of patterns, trends, and anomalies
- Assessment of data quality issues

[10 pts] Data Preparation:

- Handle missing values with justified strategies
- Address outliers appropriately
- Feature engineering (if applicable)
- Data transformations as needed
- Train-test split (if using predictive models)

[15 pts] Primary Analysis:

Choose appropriate analytical approach(es) based on your research question:

Statistical Analysis / Machine Learning / Deep learning

Part 4: Results and Interpretation (20 points)

[10 pts] Results Presentation:

- Clear presentation of findings with appropriate tables and figures
- Statistical significance and effect sizes reported
- All visualizations properly labeled and referenced
- Results organized logically to address research question

[10 pts] Interpretation and Discussion:

- Explain what results mean in practical terms
- Connect findings to literature and real-world context
- Discuss unexpected findings
- Address alternative explanations
- Acknowledge limitations and potential biases

Part 5: Research Paper (20 points)

Paper Structure (3000-4000 words):

1. Abstract (150-250 words) [3 pts]

- Research question and motivation
- Methods summary
- Key findings
- Main conclusions

2. Introduction [2 pts]

- Background and context
- Research question and objectives
- Significance of the study
- Paper organization

3. Literature Review [2 pts]

- Relevant prior research
- Theoretical framework
- Research gap addressed

4. Data and Methodology [3 pts]

- Dataset description and source
- Variables and measurements
- Analytical approach and justification
- Tools and techniques used

5. Results [4 pts]

- Comprehensive presentation of findings

- Tables and figures with clear captions
- Statistical evidence for claims

6. Discussion [4 pts]

- Interpretation of results
- Comparison with existing literature
- Practical implications
- Limitations

7. Conclusion [2 pt]

- Summary of key findings
- Contributions of the research
- Recommendations
- Future research directions

8. References

- Minimum 10 credible sources
- Consistent citation format

Writing a Research Paper

Your PDF report will be written as a formal research paper using **Overleaf**. We will provide you with the **AAAI 2025 conference template** to ensure your paper meets professional academic standards.

Find template here: [template](#)

Dataset Reference Table

Your dataset is the same as your groups from Assignments I and II:

#	Dataset Category	Specific Dataset Type/Content
1	Tourism and Transport	Tourist/visitor arrivals and tourism expenditure
2	Communication	Internet usage
3	Environment	Water supply and sanitation coverage
4	Environment	Land
5	Finance	Exchange rates
6	Science and Technology	Patents

#	Dataset Category	Specific Dataset Type/Content
7	Nutrition and Health	Health Personnel
8	Crime	Intentional homicides and other crimes
9	Energy	Energy production, trade and consumption
10	Population	Population growth, fertility, life expectancy and mortality

Here: <https://data.un.org/>

Tips for Success

- **Start Early:** Register your topic by November 28th to secure your unique angle
- **Coordinate:** Discuss with group members early to avoid overlap
- **Be Specific:** A focused, well-executed analysis is better than a broad, shallow one
- **Tell a Story:** Your analysis should have a clear narrative from question to conclusion
- **Justify Decisions:** Every methodological choice should have a clear rationale
- **Iterate:** Test multiple approaches and document what works and what doesn't
- **Proofread:** Review your paper for clarity, coherence, and technical accuracy
- **Cite Sources:** Support your methodology with references to course materials or research papers

Questions? Post on Piazza or attend office hours.

Good luck with your individual analysis!